

SCHOOL OF CHEMICAL & MATERIALS ENGINEERING
NATIONAL UNIVERSITY OF SCIENCES & TECHNOLOGY, ISLAMABAD

Paper Type: Midterm Exam **Time Allowed:** 2 hrs **Total Marks:** 100

Full Name: _____ Reg #: _____

Instructions: Attempt all questions. Write neatly and clearly in the provided space.

Section A: Objective (20 × 1 = 20 Marks)

A1. Which of the following Python data types is immutable?

- (A) List (B) Tuple
(C) Dictionary (D) Set

A2. The command `import pandas as pd` is used to:

- (A) Load plotting functions
(B) Import a data analysis library
(C) Define a variable
(D) Run an SQL query

A3. Which of the following Python features makes it suitable for data analysis?

- (A) Low-level memory management (B) Built-in support for statistical models
(C) Large ecosystem of libraries (D) Direct hardware programming

A4. What will be printed by `print(2 * 'Data' + 'Science')`?

- (A) DataScience2 (B) DataDataScience
(C) Data2Science (D) Error

A5. In Python, Jupyter Notebook is mainly used for:

- (A) Writing compiled programs (B) Interactive code and documentation
(C) Database management (D) Creating executables

A6. Which of the following is the correct order of operations in Python?

- (A) Addition, Multiplication, Parentheses (B) Parentheses, Exponentiation, Multiplication/Division, Addition/Subtraction
- (C) Multiplication, Division, Addition, Parentheses (D) Parentheses, Addition, Multiplication

A7. Which library is mainly used for machine learning tasks?

- (A) NumPy (B) Pandas
(C) Matplotlib (D) Scikit-learn

A8. In Python, the keyword used to define a function is:

- (A) func
(B) define
(C) def
(D) function

A9. What is the type of the variable created by `x = 5.0`?

- (A) int (B) float
(C) str (D) bool

A10. Which of the following statements assigns multiple variables in one line?

- (A) `x = 5; y = 10`
- (B) `x, y = 5, 10`
- (C) `x := 5; y := 10`
- (D) `x & y = 5, 10`

A11. In Python, what does a traceback indicate?

- (A) A successful function execution
- (B) A list of variable types
- (C) An error and where it occurred
- (D) A library import

A12. Which of the following will correctly print text and a variable?

- (A) `print('Value is', x)`
- (B) `print('Value is' + x)`
- (C) `print('Value is' - x)`
- (D) `print('Value is' * x)`

A13. The Python `if` statement is used to:

- (A) Define a loop
- (B) Perform conditional execution
- (C) Create a variable
- (D) Import libraries

A14. Which of the following demonstrates chained comparison?

- (A) `x > 5 < 10`
- (B) `x > 5 and < 10`
- (C) `5 < x > 10`
- (D) `x = 5 < 10`

A15. What is the result of `7 // 3` in Python?

- (A) 2.33
- (B) 2
- (C) 1
- (D) 3

A16. What is the result of `7 / 3` in Python?

- (A) 2.33
- (B) 2
- (C) 2.0
- (D) 3

A17. Which Python data type can store heterogeneous elements?

- (A) List
- (B) Tuple
- (C) Set
- (D) All of the above

A18. Which library is mainly used for numerical computations?

- (A) NumPy
- (B) Pandas
- (C) Matplotlib
- (D) Seaborn

A19. Which of the following is the correct syntax for an if-else statement?

- (A) `if x > 5: print(x) else: print(0)`
- (B) `if x > 5 print(x) else print(0)`
- (C) `if x > 5: print(x) \else: print(0)`
- (D) `if x > 5: print(x)`

A20. Which of the following statements about Python tuples is correct?

- (A) Tuples are mutable and can be changed after creation
- (B) Tuples can contain elements of different data types
- (C) Tuples do not allow duplicate elements
- (D) Tuples have no indexing support

Section B (CLO-1, C2): Code Tracing and Output ($15 \times 2 = 30$ Marks)

B1. Comprehend the following code by creating the variables $x = 2$ and $y = 3$, then determine what each of the statements below will display:

```
print('x =', x)
print('Value of', x, '+', x, 'is', (x + x))
print('x =')
print((x + y), 'x =', (y + x))
```

B2. Predict the output of the following program and explain its pattern:

```
for row in range(10):
    for column in range(10):
        print('<' if row % 2 == 1 else '>', end='')
    print()
```

B3. Fill in the missing code below so that the program prints two rows, each containing seven @ symbols. **Rewrite** the nested loops by providing the correct ranges:

```
for ***:
    for ***:
        print('@', end='')
    print()
```

B4. What does the following function do if passed the list $[1, 2, 3, 4, 5]$? **Interpret** the code logic:

```
def mystery(x):
    y = 0
    for value in x:
        y += value ** 2
    return y
```

B5. Distinguish whether the following function determines if a sequence is sorted:

```
def mystery(sequence):
    return sequence == sorted(sequence)
```

B6. Replace the `***` in the following code so that a list of heights in inches is converted into tuples with values in meters (1 inch = 0.0254 meters). **Generalize** the pattern:

```
[*** for x in [69, 77, 54]]  
list(map(lambda ***, [69, 77, 54]))
```

B7. Summarize the output of this nested loop that prints a multiplication table:

```
for i in range(1, 4):  
    for j in range(1, 4):  
        print(i*j, end=' ')  
    print()
```

B8. Translate the logic of the following function and determine its output:

```
def filter_even(lst):  
    return [x for x in lst if x % 2 == 0]  
filter_even([1, 2, 3, 4, 5, 6])
```

B9. Explain the formatted print statement below and give an example of its output:

```
name = 'Alice'  
score = 95  
print(f"Name: {name}, Score: {score}")
```

B10. Estimate the output of the following nested list comprehension:

```
matrix = [[i*j for j in range(3)] for i in range(3)]  
print(matrix)
```

B11. What will the following code print? **Infer** the boolean logic and justify:

```
x = 5
y = 10
z = 7
print(x < z < y)
```

B12. Interpret the output of the following slicing operations:

```
nums = [1, 2, 3, 4, 5]
print(nums[1:4])
print(nums[:3])
print(nums[::-2])
```

B13. Explain the result of the following boolean expression:

```
a = True
b = False
print(a and not b or b)
```

B14. Demonstrate what the following code displays:

```
s = 'Py'
print(s*3 + 'thon')
```

B15. Infer the output of this nested loop snippet:

```
for i in range(2):
    for j in range(3):
        print(i, j)
```

Section C (CLO-1, C2): Debugging and Short Questions ($15 \times 2 = 30$ Marks)

C1. Comprehend the following code intended to calculate the average of a list, and identify its errors:

```
values = [10, 20, 30, 40]
sum_val = 0
for i in range(len(values)):
    sum_val = sum_val + values[i]
avg = sum_val / len(values)
print(avg)
```

C2. Estimate a Python code to read five numbers from a user and determine their maximum and minimum without using the built-in `max()` or `min()`.

C3. Distinguish the problem in this tuple assignment:

```
day, high_temperature = ('Monday', 87, 65)
```

C4. Predict the error in this code snippet:

```
numbers = [1, 2, 3, 4, 5]
numbers[10]
```

C5. Interpret the problem in the following string assignment:

```
name = 'amanda'
name[0] = 'A'
```

C6. Explain what is wrong here:

```
numbers = [1, 2, 3, 4, 5]  
numbers[3.4]
```

C7. Identify the error in this tuple operation:

```
student_tuple = ('Amanda', 'Blue', [98, 75, 87])  
student_tuple[0] = 'Ariana'
```

C8. Infer what is wrong with the following concatenation:

```
('Monday', 87, 65) + ('Tuesday',)
```

C9. Explain the problem in this augmented assignment:

```
a = 'A'  
a += 'B'
```

C10. Interpret what happens when deleting a variable:

```
x = 7  
del x  
print(x)
```

C11. Distinguish the error when searching for a missing element in a list:

```
numbers = [1, 2, 3, 4, 5]
numbers.index(10)
```

C12. Comprehend what is wrong with this extend operation:

```
numbers = [1, 2, 3, 4, 5]
numbers.extend([6, 7, 8])
```

C13. Explain the error when removing a non-existent element:

```
numbers = [1, 2, 3, 4, 5]
numbers.remove(10)
```

C14. Predict the error that occurs when popping from an empty list:

```
values = []
values.pop()
```

C15. Translate the following cube function and identify what is missing:

```
def cube(x):
    """Calculate the cube of x."""
    return x ** 3
print('The cube of 2 is', cube(2))
```


Section D (CLO-1, C2): Conceptual Reasoning ($2 \times 10 = 20$ Marks)

D1. Given the list of numbers:

[6, 3, 5, 3, 2, 6, 4, 1, 5, 2, 7]

Summarize the steps you would take to perform the following tasks on paper:

1. Remove all duplicate elements while preserving the original order.
2. Sort the resulting list in ascending order.
3. Determine whether the final list forms a sequence of consecutive integers. If not, list the missing values.

D2. You are given the list:

[12, 7, 5, 20, 15, 8, 3]

Translate the following into paper-solvable steps:

1. Write a lambda function (or describe its logic) to select only numbers divisible by 5.
2. Use the lambda function with the concept of `filter()` to create a new list of these numbers.
3. Sort the resulting list in descending order and write it down.

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